the big hole should pulsate in and out, just as the surface of the Earth pulsates up and down after an earthquake, and those pulsations should produce gravitational waves—ripples in the curvature of spacetime that propagate out through the Universe, carrying a symphonic description of the hole.

Perhaps the greatest surprise to emerge from the golden age was general relativity's insistence that all the properties of a black hole are precisely predictable from just three numbers: the hole's mass, its rate of spin, and its electric charge. From those three numbers, if one is sufficiently clever at mathematics, one should be able to compute, for example, the shape of the hole's horizon, the strength of its gravitational pull, the details of the swirl of spacetime around it, and its frequencies of pulsation. Many of these properties were known by 1975, but not all. To compute and thereby learn all the remaining black-hole properties was a difficult challenge, precisely the kind of challenge that Chandrasekhar loved. He took it up, in 1975, as his personal quest.

For nearly forty years, the pain of his 1930s battles with Eddington had smoldered inside Chandrasekhar, impeding him from a return to research on the black-hole fates of massive stars. In those forty years he had laid many of the foundations for modern astrophysics—foundations for the theories of stars and their pulsations, of galaxies, of interstellar gas clouds, and much more. But throughout it all, the fascination of the fates of massive stars had attracted him. Finally, in the golden age, he had overcome his pain and returned.

He returned to a family of researchers who were almost all students and postdocs. The golden age was dominated by youth, and Chandrasekhar, young at heart but middle-aged and conservative in demeanor, was welcomed into their midst. On extended visits to Caltech and Cambridge, he could often be seen in cafeterias, surrounded by brightly and informally bedecked graduate students but himself attired in a conservative dark gray suit—"Chandrasekhar gray" his youthful friends called its color.

The golden age was brief. Caltech graduate student Bill Press had given the golden age its name, and in the summer of 1975, just as Chandrasekhar was embarking on his quest to compute the properties of black holes, Press organized its funeral: a four-day conference at Princeton University to which only researchers under the age of thirty were invited.[1] At the conference, Press and many of his young colleagues agreed that now was